

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – ARCHITECTURE / MODELING (SECURE INFRASTRUCTURE)

Course Code:	CSA5803	Course Title:	DEVSECOPS ENGINEERING AND APPLICATION SECURITY
CO Assessed:	CO4 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 2
Weightage:	30% of CIA	Total Marks:	100
CO4 Statement:	Design secure DevSecOps infrastructure by developing architecture diagrams, modelling deployment workflows, Identity and Access Management (IAM) relationships, and GitOps workflows.		
Student Name:	ADHITHYAN S	Reg. No.:	192421102
Section / Batch:	B.TECH IT	Date:	

Instructions to Students

- Design all architecture diagrams, deployment workflows, IAM relationships, and GitOps workflows originally and clearly.
- Use DiagrammeR or Draw.io for architecture modelling and maintain consistent labels, trust boundaries, data flows, and security controls.
- Clearly represent IAM roles/permissions, Infrastructure as Code (IaC), GitOps flow, and DevSecOps security integration wherever applicable.
- Include editable source files for diagrams, IaC/configuration files, methodology documentation, and all supporting project artifacts.
- Submit the GitHub repository link and final PDF report through Google Classroom before the specified deadline; verify that the repository is accessible and complete.

Question Type	Focus Area	Questions	Marks
ARCHITECTURE / MODELING	Design and Application Based	Q1	Q1 – 100
GRAND TOTAL:		100 Marks	

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
IAM Model and Access Control Design	20%	Designs a comprehensive IAM model with clearly defined identities, roles, permissions, authentication and least-privilege access controls	Designs an appropriate IAM model with most roles and access controls correctly represented and only minor gaps	Provides a basic IAM model with limited role definition, permissions or access-control details	Provides an incomplete or inaccurate IAM model with weak or inappropriate access controls
Infrastructure as Code (IaC) Modelling	30%	Accurately models infrastructure provisioning through IaC with reusable, consistent and secure configuration components	Models IaC workflow correctly with appropriate configuration structure and only minor omissions	Provides a basic IaC model with limited automation, structure or security considerations	Provides an incomplete or incorrect IaC model with poor configuration and automation representation
GitOps Workflow and Version Control	20%	Clearly models a complete GitOps workflow including repository changes, review, automated synchronization, deployment and desired-state management	Models the GitOps workflow correctly with most stages represented and minor workflow gaps	Provides a basic GitOps workflow with missing stages or unclear repository-to-deployment relationships	Provides an incomplete or inaccurate GitOps workflow with weak version-control integration
Security Integration and Workflow Automation	20%	Effectively integrates IAM, IaC and GitOps with security validation, approvals, automation and auditable workflow controls	Integrates appropriate security and automation controls with only minor gaps	Shows basic security integration and automation but several controls or workflow relationships are missing	Shows limited or incorrect integration of security controls and workflow automation
Documentation and Workflow Clarity	10%	Clearly documents IAM, IaC and GitOps components and presents the complete workflow with strong technical justification	Provides clear documentation and workflow explanation with minor clarity or justification gaps	Provides basic documentation with limited explanation of workflow components and relationships	Provides incomplete or unclear documentation with little technical justification

Rubrics:

Criteria	Weightage	Q5 (10)	Total Marks (50)
IAM Model and Access Control Design	25%		
Infrastructure as Code (IaC) Modelling	30%		
GitOps Workflow and Version Control	20%		
Security Integration and Workflow Automation	15%		
Documentation and Workflow Clarity	10%		
Total per Question	100 Marks		

100

DevSecOps GitOps Secure Architecture

